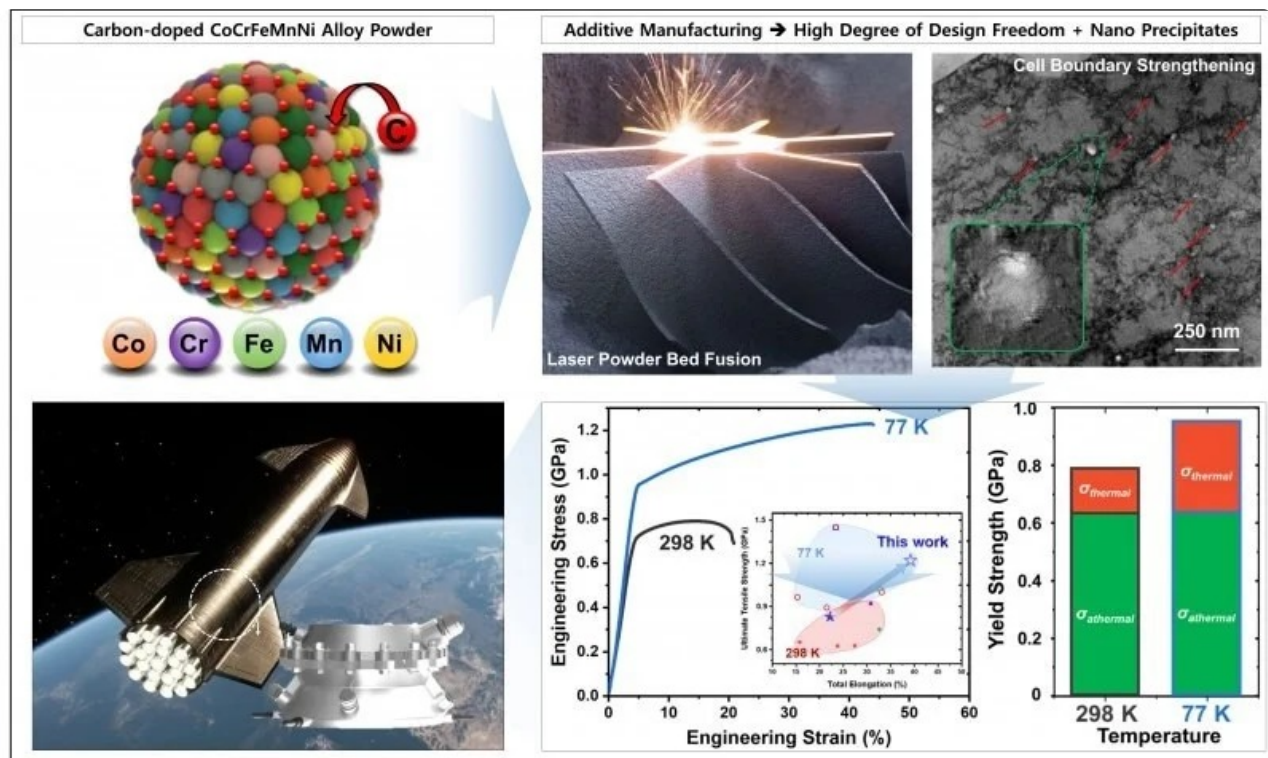


3D-Printed Alloy Paves the Way for Extreme Environment Applications

Reviewed by Lexie Corner

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A team of researchers from the Korea Institute of Materials Science (KIMS), in collaboration with Gyeongsang National University and the Pohang University of Science and Technology (POSTECH), has developed a high-performance metal 3D-printed alloy specifically designed for space environments. The newly created alloy demonstrated exceptional mechanical properties at cryogenic temperatures as low as -196°C , highlighting its potential for use in extreme environments, including space travel. The study was published in *Additive Manufacturing*.



The 3D printing technology for designing structural materials having outstanding cryogenic performance developed by the research team for space exploration applications. It boasts superior combination of strength-ductility in cryogenic temperature, and can be tailored by controlling microstructure and process parameters for desired application. Image Credit: Korea Institute of Materials Science

Led by Dr. Jeong Min Park from the Nano Materials Research Division at KIMS, with contributions from Professor Jung Gi Kim of Gyeongsang National University and Professor Hyung Seop Kim of POSTECH, the team enhanced the cryogenic properties of a

CoCrFeMnNi alloy by adding a small amount of carbon. The alloy powder was then prepared using Laser Powder Bed Fusion (LPBF) and Metal [Additive Manufacturing](#) processes, also known as Metal [3D Printing](#).

The method enhances the alloy's strength by dispersing nano-carbides along the edges of nano-sized cell structures, maximizing the impact of carbon addition. This approach enabled the team to achieve a combination of tensile strength (resistance to stress) and ductility (ability to deform without breaking) in cryogenic conditions that surpasses carbon-free alloys by over 140 %. Specifically, the alloy exhibits twice as much elongation at 77 K as it does at 298 K.

Additionally, this method provides a possible guideline for alloying design in additive manufacturing, resulting in high-performance products with exceptional load-bearing capacity for cryogenic applications. This technology's capacity to fine-tune microstructure via additive manufacturing is another important differentiator.

This technology can produce complex components such as turbine nozzles, which harvest energy, and fuel injectors for space exploration rockets, improving the longevity and performance of parts in extreme environments. It also overcomes the limitations of existing 3D-printed metals, which often lack toughness at low temperatures.

“ *This research presents a significant breakthrough in developing new alloys for extreme environments, offering new possibilities. Through 3D printing technology that surpasses the manufacturing limits of conventional space exploration components, we can significantly improve the performance of parts used in space launch vehicles.*

*Dr. Jeong Min Park, Study Senior Researcher and Project Leader, Korea
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The primary KIMS projects, “Development of design for additive manufacturing to develop superhard heterogeneous materials with complex design,” and “Development of High Performance Materials and Processes for Metal 3D Printing,” supported the study. The study team intends to carry out more experiments to confirm the technology's performance in harsh settings and increase its commercialization potential.

Journal Reference:

Park, H., *et al.* (2024) Cryogenic tensile behavior of carbon-doped CoCrFeMnNi high-entropy alloys additively manufactured by laser powder bed fusion. *Additive Manufacturing*. doi.org/10.1016/j.addma.2024.104223.

Source:

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